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Assignment Research: Report 4 Ransomware Recovery

What did you do?

I developed a ransomware recovery plan for my home network, leveraging prior research and the four-phase recovery framework outlined by Dzimielia and Jennex in their account of the 2019 Texas municipal ransomware attacks. While their work addressed large-scale municipal networks, I adapted the principles to fit the scale and resources of a home environment.

First, I conducted an asset inventory and prioritization using Nessus, Nmap-Zenmap, and manual verification. Like the Texas National Guard team described in the case study, I created a tiered importance list to identify mission-critical devices before moving to less critical IoT devices. Second, I assessed vulnerabilities using Nessus and Shields UP!. In 2019, many municipalities were unprepared, with outdated systems and weak or missing password policies. My network, by contrast, has current firmware and strong password controls. Still, the identification of closed-but-responsive ports on my router reflected the type of external exposure attackers exploited in the municipal case, reinforcing the need to harden those settings.

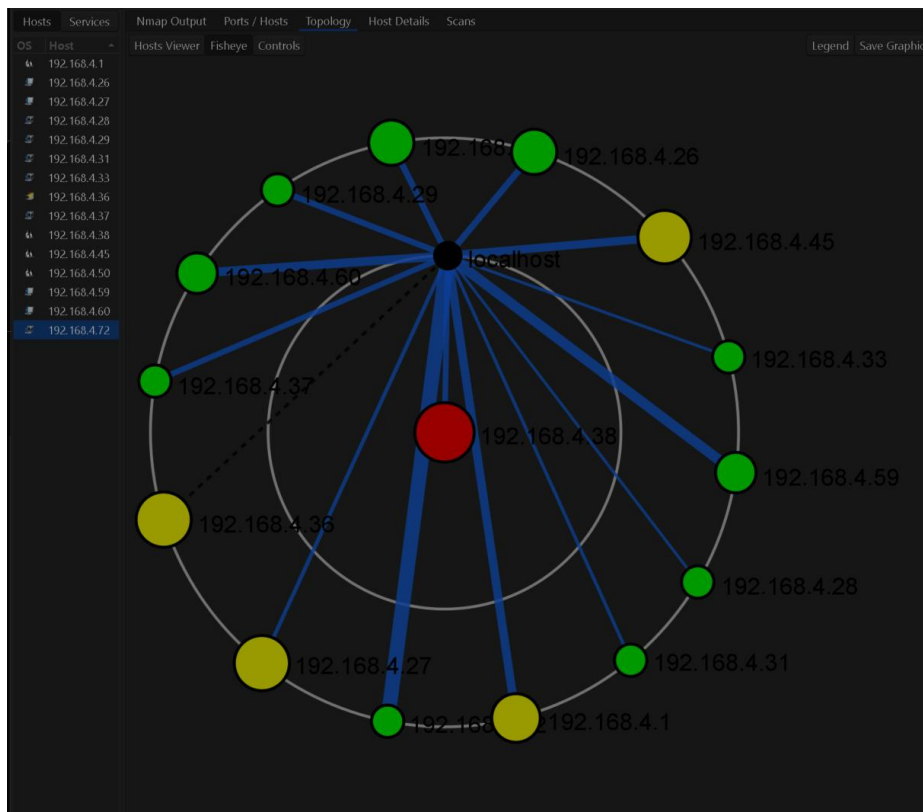
Third, I implemented a backup and credential security strategy. Unlike the municipalities that lacked structured backups, I maintain both cloud-based (OneDrive, Google Drive) and offline backups, updated quarterly. Past data-loss incidents reinforced the importance of keeping backups, so this is now a non-negotiable part of my security strategy. I developed secure, unique passwords (over 250), set up seven passkeys, and securely recorded my Windows license key

(redacted). Fourth, I checked for system patches and updates. In the municipal case, out-of-date servers and a lack of patching facilitated the spread of ransomware. My plan includes enabling automatic updates and scheduling regular Nessus scans to maintain an up-to-date, monitored environment.

Finally, I formalized a ransomware recovery playbook. Following the study's model, my plan specifies: (1) rapid restoration of critical assets; (2) immediate vulnerability remediation; (3) systematic restoration of all remaining components; and (4) structured reflection and recovery plan updates. Aligning with a proven real-world framework keeps the plan practical and evidence-based.

What are the results?

These efforts produced a clear understanding of my home network's assets, vulnerabilities, and recovery readiness. Key deliverables include a prioritized device inventory, a backup regimen for critical data, a list of identified weaknesses with mitigation steps, and a formal ransomware recovery plan. These deliverables not only document my current security posture but also guide how to respond effectively to a ransomware incident. Below is a breakdown of the results:



Asset Inventory:

Device Name	Type	Mobile	Importance
Eero	Router	No	Critical
iPhone 12 Mini	Phone	Yes	Low
Google Home	IoT	No	Low
Wyze Camera 1	IoT	No	Low
Wyze Camera 2	IoT	No	Low
Wyze Camera 3	IoT	No	Low
Wyze Camera 4	IoT	No	Low

Wyze Camera 5	IoT	No	Low
Samsung TV	TV	No	Low
Vizio TV	TV	No	Low
Eero Mesh	Extender	No	Low
Feit Bulb 1	IoT	No	Low
Feit Bulb 2	IoT	No	Low
Galaxy Tab A9+	Tablet	Yes	Low
Personal PC (redacted)	Desktop	No	High/Critical

Table: Home network devices and their importance. “Critical” assets cannot be down for more than a day without serious impact; “High” indicates intolerable downtime beyond ~2-5 days; “Medium” means a loss of more than two weeks would cause some disruption; “Low” means minimal impact even if lost indefinitely.

From this inventory, I identified that the Eero router is my only truly critical device, while my primary computer (“Personal PC (redacted)”) is of high importance and nearly critical to daily productivity. Most other devices are low priority and would not significantly hinder operations if unavailable for an extended time. This prioritized list ensures that in a recovery scenario, I know exactly where to focus first.

Data Backup and Password Management: I confirmed that important files and photos are backed up to Microsoft OneDrive, with a few additional files on Google Drive. For data not already in

the cloud, I created local backup copies (for instance, an external drive image of my PC).

Backups are scheduled quarterly, and for critical data, I may increase the backup frequency to reduce potential loss. In terms of credentials, I compiled a list of all passwords and access keys. All of these are now stored securely (in encrypted files) so that I can rebuild quickly. A planned improvement is migrating to a dedicated password manager for stronger security and convenience.

Images from previous research reports below:

Identified Vulnerabilities and Mitigations: Scanning revealed weaknesses that I addressed or plan to address. Shields UP! reported that 72 ports were in a Closed state but still responded to probes. This exposure could give attackers information, so I intend to adjust my firewall settings to Stealth those ports (making them completely invisible externally). Nessus identified one Medium-severity vulnerability on an uncommon port (an SSL-related warning on port 51192). I mitigated this and secured the service. With many IoT devices on the LAN, I hardened them by changing default credentials and keeping firmware up to date. All systems are now patched and updated to the latest versions.

To further strengthen my security, I configured recurring Nessus scans and enabled automatic updates on all devices. I will periodically rerun external port scans (e.g., ShieldsUP!) to validate the network's outward-facing profile. I am also evaluating a premium anti-virus/anti-malware tool to complement Microsoft Defender.

Ransomware Recovery Plan: All the information gathered above is fed into a formal ransomware recovery plan, which I can follow in the event of a ransomware attack or similar disaster. The plan is structured in four phases, adapted from best-practice frameworks (such as the approach discussed by Dzimiela and Jennex). The steps are as follows:

1. Identify and restore mission-critical assets: If an incident occurs, immediately identify the most critical assets (hardware, software, and data) that are needed to resume basic operations. These must be restored first. If any essential data or system is encrypted or otherwise inaccessible, retrieve the latest clean backup and restore that data onto the device (or a replacement device). In a worst-case scenario where a device is completely compromised, prepare to reimage or replace the device and then load it with data from backups.
2. Patch or remediate vulnerabilities: Investigate how the ransomware breached the system – whether it exploited a known vulnerability or a new one. Any identified vulnerability will be patched or fixed as soon as possible. If the exact cause of the breach isn't apparent, conduct a deeper analysis using logs and basic forensics to find any hidden weaknesses.
3. Clean up and restore remaining components: Proceed to restore all other systems and data that were not covered in the first phase. This includes recovering less critical devices and files, reinstalling software, and bringing the entire network to normal operation. During this cleanup, remove any traces of malware or malicious changes left by the ransomware.
4. Reflect and strengthen defenses: Analyze why and how the incident happened and document the lessons learned. This may involve actions such as upgrading to more secure software or hardware, reconfiguring network settings, and providing additional security

training to network users. Also, review and update security policies and procedures.

Determine whether the attack was specifically targeted or a broad, opportunistic ransomware campaign. Any changes or improvements identified here will be implemented going forward, making the recovery plan a living document that evolves with each insight.

What did you learn?

Through this research assignment, I learned that small businesses and home users often have significant security gaps, making them attractive ransomware targets. In my previous IT help desk experience at a small business, security practices were basic, and this assignment highlighted how that is often the case. It was eye-opening to compare my network's initial state to best practices: I saw why attackers might choose victims like small businesses, who are more likely to pay a ransom since they aren't sure how else to recover. The experience underscored the importance of implementing even basic security measures in any network, no matter how small.

One of the biggest takeaways for me is the value of preparation in cybersecurity, especially backup and recovery planning. By doing this in advance, I learned that you can drastically reduce the impact of a ransomware attack. If an attacker were to encrypt my data, I now possess recent backups to restore from, which means I wouldn't be at their mercy. I also learned about the importance of password management and protecting credentials. Sorting through accounts and strengthening passwords indicated that it is easy to neglect this aspect. I intend to use a password manager moving forward, which is something I hadn't done before.

This assignment was a valuable exercise in applying what I learned to a real-world scenario. By combining knowledge from all three previous research reports and my own

experience, I was able to produce a cohesive plan. This taught me how the various pieces of cybersecurity (asset management, patch management, network monitoring, incident response planning, etc.) fit together. It's one thing to learn about these topics in isolation, but linking them in a single project gave me a much deeper understanding of how to manage an organization's security holistically. I feel much more confident in my ability to assess and improve a network's security.

This project also showed me the value of having a structured recovery plan and how it can save an organization from disaster. I learned that responding to ransomware (or any cyber incident) requires clear roles, priorities, and procedures defined ahead of time. In the future, I will carry this lesson into any IT or security role. This assignment has essentially given me a reusable playbook for ransomware recovery that I can adapt and apply elsewhere. Moreover, by reflecting on why the incident might happen and how to prevent it, I understood that cybersecurity is a continuous cycle of improvement. I also feel better equipped to help others secure their networks.

In summary, I learned not just technical skills, but also the strategic mindset of anticipating threats and being proactive. I conclude this will be immensely valuable as cyber-attacks continue to rise, and I'm ready to apply these lessons to safeguard digital assets in any environment I find myself in.